



RESEARCH ARTICLE

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Key Points:

- It is possible to locally improve on European Centre for Medium-Range Weather Forecasts spectral wave forecast by learning a correction from historical wave buoy measurements
- Improvements are significant in wave peak period, mean period, and swell significant wave height

Correspondence to:

A. Filoche,
arthur.filoche@uwa.edu.au

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Author Contributions:

Conceptualization: Jeff Hansen
Data curation: Arthur Filoche
Formal analysis: Arthur Filoche
Funding acquisition: Jeff Hansen
Investigation: Arthur Filoche, Jeff Hansen
Methodology: Arthur Filoche, Jeff Hansen
Project administration: Jeff Hansen
Resources: Arthur Filoche, Jeff Hansen
Software: Arthur Filoche
Supervision: Jeff Hansen
Validation: Arthur Filoche, Jeff Hansen
Visualization: Arthur Filoche
Writing – original draft: Arthur Filoche
Writing – review & editing: Jeff Hansen

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Site-Specific Post-Processing of Spectral Wave Forecast by Learning From Buoy Measurements

Arthur Filoche^{1,2} , Jeff Hansen¹ , Kevin Vinsen², and Travis Dawson¹ 

¹University of Western Australia, Oceans Institute, Crawley, WA, Australia, ²University of Western Australia, ICRAR, Crawley, WA, Australia

Abstract Operational wave prediction systems provide critical information across a range of marine industries and for public safety. Yet operational numerical wave forecasts can exhibit systematic biases, such as in wave height and arrival time of swell events. Focusing on a deepwater site in the Browse Basin off the northwest coast of Australia, here we described a method developed to mitigate these errors based on a site-specific post-processing supervised deep learning regression task. Using 3.5 years of historical spectral observations and numerical forecasts, we mapped the European Centre for Medium-Range Weather Forecasts (ECMWF) directional wave variance spectrum to the one-dimensional variance density spectrum recorded by a local moored wave buoy. To capture unresolved local hydrodynamics and seasonality, this forecast was augmented with specific context variables, namely local tidal water levels and temporal embeddings. The proposed architecture adapted an attention-based encoder-decoder model modulated with context-conditioning. Results demonstrated that our model significantly improved upon the ECMWF baseline in the low-frequency swell regime across a 5-day forecast horizon. Specifically, at a 120-hr forecast lead time, the model reduced the Root Mean Square Error for swell significant wave height from 0.16 to 0.12 m (a 25% reduction), and reduced swell peak period errors from 2.22 to 1.39 s (a 37% reduction). Furthermore, the architecture successfully mitigated the premature swell arrival bias characteristic of this geographic region. However, the network struggled to resolve the stochastic variance in higher-frequency wind seas.

Plain Language Summary Numerical ocean wave forecast models provide essential information for safe maritime operations. However, these models are not perfect and can contain systematic errors. For example, within the deep water Browse Basin, standard models consistently predict that large, long-distance waves (swells) will arrive earlier than they actually do. To fix these local errors, we trained an artificial intelligence (AI) system using historical data. The AI learned by comparing the standard wave forecasts against actual physical measurements taken by an ocean buoy over several years. By also feeding the AI information about the local tides and the time of year, it learned to recognize and correct the forecast's mistakes. Our results show that this localized AI correction successfully improves the forecast for ocean swells, fixing the timing of when these waves will arrive and more accurately predicting their size. However, the AI had difficulty improving the forecasts for shorter, more chaotic waves created by local winds.

1. Introduction

Operational numerical wave prediction systems, such as those run by the national forecasting agencies, provide critical information for maritime and coastal industries and public safety applications. Operational wave forecasts are produced using numerical spectral wave models that may assimilate observations (Bidlot, 2012) and are forced by operational wind and ice forecasts. Errors in the spectral wave forecasts can arise from various sources, including incomplete physics, initial and boundary conditions, coarse-grid discretization, sparse and noisy observations, as well as the choice of data assimilation algorithm. While national and commercial weather prediction centers continue to improve the accuracy of their forecasts incrementally (ECMWF, 2023, 2026), errors, sometimes large, remain. For instance, when evaluating numerical wave forecasts against in situ buoy measurements, a systematic bias appears (Sun et al., 2026), resulting in early swell arrival time prediction along the Northwest Shelf (NWS) of Australia which features many large offshore energy installations.

Where observational datasets exist, comparison between repeated forecasts and the resulting observations constitutes an opportunity to learn a statistical site-specific correction. Mitigating errors by learning a transfer function between historical forecasts and observed ground truth is not a new topic and has been practiced since

(Glahn & Lowry, 1972) introduced Model Output Statistics (MOS). It has already been used in a wave forecasting context to correct bulk parameters such as significant wave height H_s and mean wave period (Reikard et al., 2011). While successful, such frameworks exhibit limitations. Relying exclusively on H_s discards important frequency-directional information described by the wave spectrum. Also, MOS does not naturally handle multi-dimensional output, requiring independent models per forecast horizon and variable (Vannitsem & Nicolis, 2008).

Numerical weather forecasts, data-intensive by nature, have logically been enhanced by recent advances in Machine Learning (ML), at every step of the forecasting pipeline (Düben et al., 2021). More particularly for post-processing, ML systems have been shown to correct biases and phase shifts in forecasts while better quantifying uncertainties (Haupt et al., 2021). Specialized architectures like ConvMOS (Steininger et al., 2023) have demonstrated the use of Deep Learning (DL) to handle the correction of multivariate and high-dimensional forecasts. Therefore, we postulate that deep neural networks are uniquely equipped to process a physically structured 3-dimensional (frequency, direction, time) spectral wave forecast and directly regress time series observations. Additionally, the flexibility of recent architectures unlocks seamless integration of relevant contextual variables, of any modality, that may not be included within the numerical models.

In this work, we propose to leverage DL to post-process the European Centre for Medium range Weather Forecasting (ECMWF) spectral wave forecast at a specific location on the NWC of Australia. We formulate the problem as a supervised regression against the wave variance density spectrum, recorded locally by a moored directional wave buoy. To do so, we adapt an attention-based encoder-decoder architecture from (Archambault et al., 2024), proven efficient at processing spatio-temporal geoscientific data. We augment it with context-conditioning mechanisms (Perez et al., 2018) to include local environmental variables such as tides (which induce local currents that impact the waves). After optimization, the proposed model improves on ECMWF baseline, predominantly in the low-frequency swell band. Importantly, the model mitigates the premature swell arrival forecast characteristic of this geographic region.

The remainder of this article is organized as follows: Section 2 presents the considered spectral wave data set. Section 3 describes the deep architecture and the optimization method. Section 4 presents the quantitative forecast skills and spatiotemporal error decompositions, and Section 5 concludes with a critical discussion about the limitations and perspectives of the proposed method.

2. Spectral Wave Data Set

2.1. ECMWF Operational Wave Forecast

The forecasts we aim to post-process are the operational wave forecasts from ECMWF, computed using the ecWAM spectral wave model including assimilation of satellite derived H_s . The details of the operational wave model including atmospheric forcing and assimilation are described in (ECMWF, 2023). The directional spectra therefore already contain the wave model response to the forecast atmospheric forcing. However, the corresponding atmospheric wind vectors were not retrieved as separate inputs to the post-processing model. At each point of the spatial grid, the ecWAM model represents the sea state with the 2D directional wave variance spectrum, denoted $E(f, \theta)$, providing the distribution of wave variance density as a function of frequency (f) and direction (θ). Using the ECMWF Meteorological Archival and Retrieval System (MARS) archive system, the operational forecast is retrieved at 13.75°S, 123.26°E, the closest grid point to the buoy of interest. Selected forecasts were issued twice daily, and we selected a 5-day horizon with a 3-hr temporal step (40 lead times). For the period covered by this study, each archived directional spectrum contained 36 directional bins and 36 frequency bins. We downloaded the full spectra but retained only the first 29 frequency bins, spanning approximately [0.034, 0.49] Hz, because the spectral energy in the higher-frequency bins was negligible at the study site. The resulting model input therefore contains 36 directional bins and 29 frequency bins. Approximately 3.5 years of historical forecasts, ranging from 2020 to 2024 were obtained from the ECMWF MARS archive. An example of a forecast is shown in Figure 1.

2.2. Directional Wave Buoy

The observational ground truth associated with the forecasts was collected from a moored directional wave buoy, a Datawell Waverider Mk4 (Datawell BV, 2006), deployed adjacent to an offshore floating natural gas facility in a

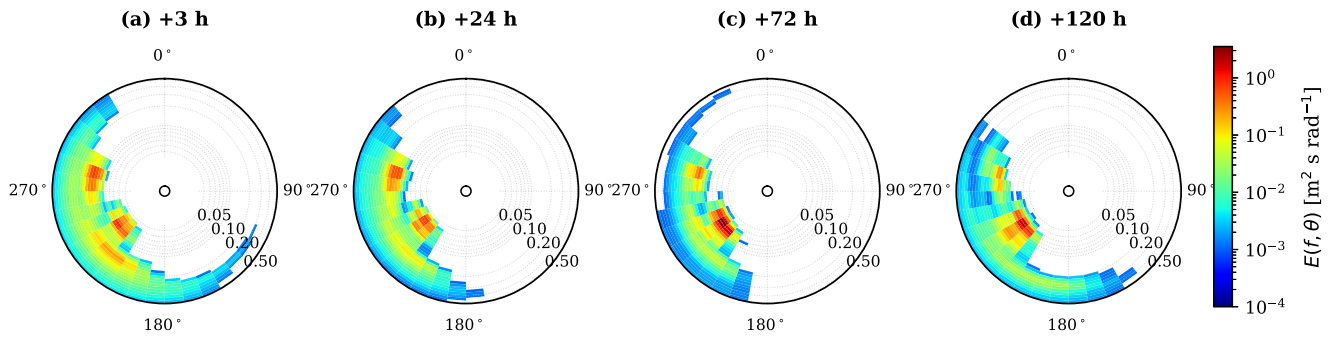


Figure 1. European Centre for Medium-Range Weather Forecasts operational wave forecast issued on 2022-03-26 at the target site. The panels display the 2D directional wave variance spectrum, $E(f, \theta)$, at four discrete forecast lead times (+3, +24, +72, and +120 hr), illustrating the temporal progression of the wave field over the 5-day predictive horizon. The data are visualized in a meteorological polar projection (“coming from” convention), where the angular axis denotes wave direction in degrees and the radial axis represents frequency over [0.034, 0.49] Hz on a logarithmic scale. Wave variance density ($\text{m}^2 \text{Hz}^{-1} \text{rad}^{-1}$) is represented using a logarithmic color scale to capture both energetic local wind seas and lower-frequency remote swells. Missing or near-zero variance-density bins (10^{-4}) are masked for clarity.

water depth of approximately 250 m. The buoy measures three-dimensional displacement and, after band-pass filtering, estimates the motion of a particle at the sea surface. Buoy displacements are recorded for 30-min intervals at a sampling rate of 2.56 Hz. From these records, it is possible to estimate the first-5 directional Fourier coefficients of the wave spectrum (Kuik et al., 1988) classically denoted $[a_0, a_1, b_1, a_2, b_2](f) = \int_0^{2\pi} E(f, \theta) [1, \cos(\theta), \sin(\theta), \cos(2\theta), \sin(2\theta)] d\theta$. In this work, since $[a_1, b_1, a_2, b_2]$ are noisier and less interpretable, we decide to focus on correcting the one-dimensional variance-density spectrum, $a_0(f) = E(f)$, as the target variable for the post-processing, hence discarding the correction for directional distribution. Observations were filtered to remove unrealistic records using thresholding for low significant wave heights and high peak periods. Each buoy timestamp denotes the start of the corresponding 30-min acquisition window.

In Figure 2 we display the climatology of the site of interest as recorded by the wave buoy between 2020 and 2024. Two wave systems prevail: low-frequency swell from the Southern Ocean and a low-frequency wind sea centered around 0.06 and 0.13 Hz, respectively. As established by global climatologies (Hanley et al., 2010), this deep water region operates predominantly within a swell-dominated regime. Intense mid-latitude storms in the Southern Ocean generate long-period waves that propagate northeastward across the Indian Ocean. At the target site, this low-frequency energy arrives consistently from a narrow southwest directional band, peaking in intensity during the austral winter when Southern Ocean storm activity is at its maximum. Conversely, the higher-frequency wind sea is directionally spread and distributed across two primary modes. This bimodal

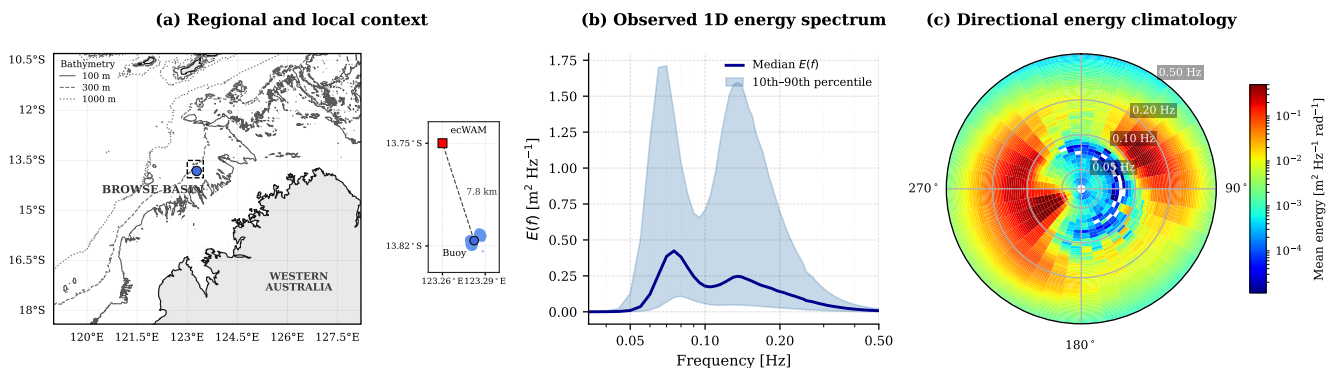


Figure 2. Spectral climatology and measurement context for the study site. (a) Regional bathymetry of northwest Australia and the Browse Basin, together with a local panel showing the measured buoy track and the nearest native reduced-grid ecWAM node at 13.75°S, 123.26°E. The single-point Meteorological Archival and Retrieval System request selected the closest native node without horizontal interpolation. (b) Median observed one-dimensional energy-density spectrum, $E(f)$; shading shows the frequency-wise 10th–90th percentile range. (c) Variance-weighted mean-direction climatology, obtained by assigning the observed one-dimensional energy at each frequency to its corresponding first-moment mean-direction bin and averaging over time. This quantity is not a reconstruction of the complete directional spectrum $E(f, \theta)$.

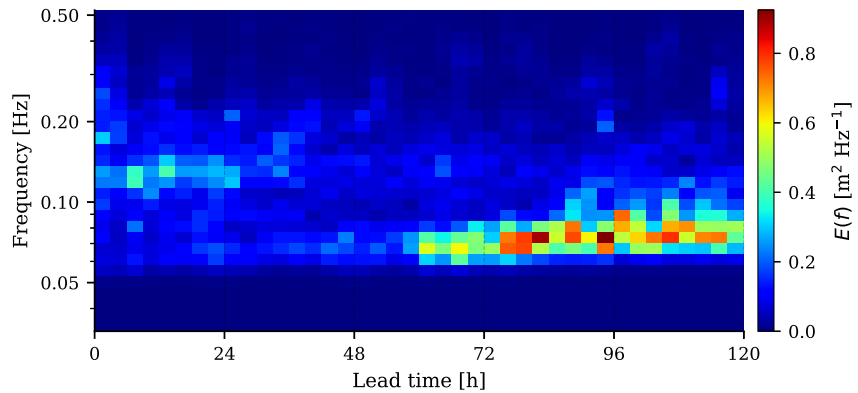


Figure 3. Temporal evolution of the target (interpolated) 1D wave variance spectrum $\hat{E}(f)$, derived from in situ Datawell Waverider Mk4 buoy measurements at the target site. This sequence serves as the ground-truth regression target for the post-processing model across the 5-day (+120 hr) forecast horizon. The heatmap displays wave variance density (m^2/Hz) as a function of discrete 3-hourly forecast lead times (x-axis) and logarithmically spaced frequency bins spanning $[0.034, 0.51]$ Hz (y-axis). Strict temporal alignment is enforced between the buoy records and the numerical forecast issuances.

distribution is driven by the region's seasonal wind oscillation: the Southeast Trade Winds generate northwest-propagating seas during the austral winter, while the Northwest Monsoon forces southeast-propagating seas during the austral summer.

2.2.1. Spectral and Temporal Harmonization

2.2.1.1. Time Alignment

The 30-min interval buoy observations are selected at times matching the 3-hourly resolution of the ECMWF forecast (e.g., 00:00, 03:00, 06:00 UTC), discarding intermediary timestamps.

2.2.1.2. Frequency Regridding

For the sake of simplicity, we harmonize the dataset on the coarser ECMWF frequency grid. To do so, the buoy observations are projected onto the target grid using piecewise-linear interpolation. Because interpolation may modify the integrated spectral variance, the interpolated spectrum is rescaled to preserve the zeroth spectral moment, $m_0 = \int E(f) df$, and therefore the corresponding significant wave height, $H_s = 4\sqrt{m_0}$. This ensures consistency with the native buoy observations. The processing is formalized in Equation 1, where $E_{\text{interp}}(f)$ is the final rescaled variance density at the target ECMWF frequency f , and E_{raw} is the native buoy variance density over the original frequency grid f_{raw} . The operator $L(\cdot)$ denotes the piecewise-linear interpolation function onto the target frequency grid f_{interp} , and α is the normalization scalar ensuring conservation of m_0 . For the interpolation step, f_i and f_{i+1} represent the adjacent frequencies on the native buoy grid that bound the target frequency f , with E_i and E_{i+1} being their corresponding raw variance-density values.

$$E_{\text{interp}}(f) = \alpha L(E_{\text{raw}})(f) \quad \text{w.r.t.} \quad \alpha = \frac{\int E_{\text{raw}} df_{\text{raw}}}{\int L(E_{\text{raw}}) df_{\text{interp}}} \quad (1)$$

$$L(E_{\text{raw}})(f) = E_i + \frac{f - f_i}{f_{i+1} - f_i} (E_{i+1} - E_i), \quad f \in [f_i, f_{i+1}]$$

In Figure 3 we display an example of a processed buoy observation, aligned with the forecast in Figure 1, that will act as ground truth in the following experiments.

2.3. Context Variables

Seeking to enhance the performance of the post-processing, we investigate the relevance of contextual variables such as tidal elevation, the date (e.g., to capture seasonal variability) and previous buoy measurements.

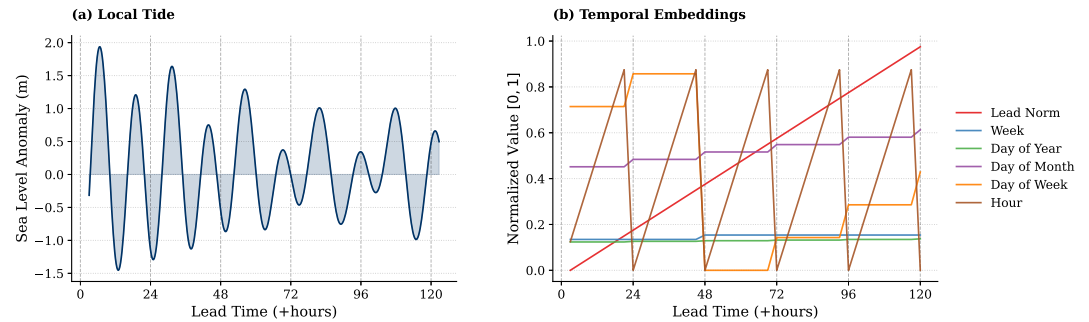


Figure 4. Context variables. (a) Local sea level anomalies (m) derived from harmonic tidal analysis, provided at 15-min intervals to capture high-frequency hydrodynamic forcing unresolved by the coarse 0.1° numerical wave grid. (b) Temporal coordinates normalized strictly to a continuous $[0, 1]$ range, comprising Lead Time, Week, Day of Year, Day of Month, Day of Week, and Hour.

2.3.1. Tidal Conditions

In the operational ECMWF coupled forecasting system considered here, the ocean circulation model NEMO is implemented without tides. Consequently, the ecWAM wave model is not forced by tidal currents or time-varying water depths. This is particularly relevant in the studied area, where large (~ 5 m range) barotropic tides interact with steep bathymetry to generate intense, highly localized currents and internal waves that can affect the sea state even at water depths exceeding 100 m (Rayson et al., 2011). To provide the post-processing model with information related to these processes, local tidal elevation is included as a physically motivated proxy for the phase and magnitude of the unresolved tidal hydrodynamics. It should be noted that this proxy does not explicitly represent the associated tidal currents. Tide series are estimated using harmonic analysis, where historical water-level records are decomposed into a summation of discrete cosine waves. Since these astronomical frequencies are deterministic, tidal water levels can be projected throughout the study period. The input consists of a 120-hr window of tidal water levels sampled at 15-min intervals (480 samples), aligned with the temporal horizon of each forecast issuance. Panel (a) of Figure 4 displays an example of the predicted tidal water levels associated with the forecast issued on 15 February 2020 at 00:00 UTC.

2.3.2. Time Encoding

Knowledge of time may help the post-processing model to capture seasonality in the forecast errors. To represent this information, we encode the date into six variables, normalized between $[0, 1]$: forecast lead time, week of the year, day of year, day of month, day of week, and hour of the day. In the panel (b) of Figure 4 is displayed an example of such encoding for the forecast issued on 15/02/2020 at 00:00 UTC.

2.3.3. Previous Buoy Observations

To provide the model with the most recent wave conditions, a 5-day historical window of buoy observations is included. This sequence consists of 40 observations extracted at a 3-hr resolution immediately preceding the initialization time, and so matching the dimension of the ground truth target depicted in Figure 3.

3. Proposed Post-Processing Methodology

3.1. Regression Task

To mitigate systematic ECMWF forecast errors and align with local observations, the proposed post-processing methodology is, similarly to MOS, formalized as supervised regression task. The regressor g_ϕ , parameterized by ϕ , takes as input the ECMWF 2D spectral forecast tensor $\mathbf{X}$ and additional context tensor $\mathbf{C}$ to output a predicted 1D spectral variance density tensor $\hat{\mathbf{E}}$. Formally, the mapping is defined as $g_\phi : (\mathbf{X}, \mathbf{C}) \rightarrow \hat{\mathbf{E}}$.

The parameters ϕ are estimated by minimizing the expected risks over the joint distribution of the dataset $D = \{(\mathbf{X}, \mathbf{C}), \mathbf{E}\}$. Specifically, we seek to find the optimal parameters ϕ^* that satisfy:

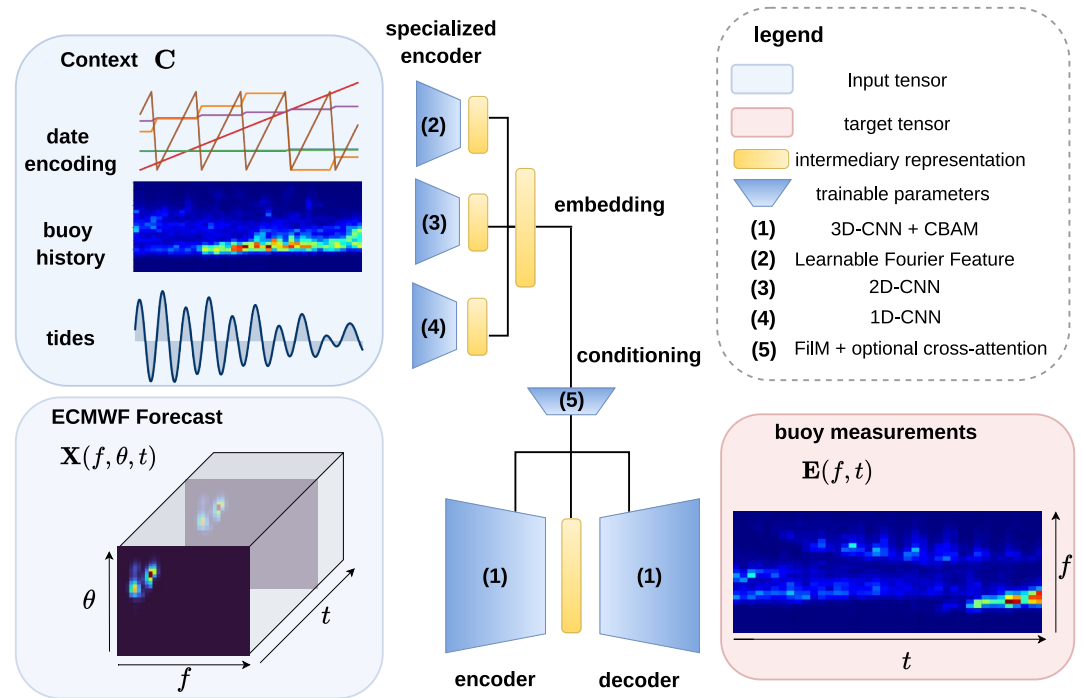


Figure 5. Architecture scheme of g_ϕ . The primary input is the 2D spectral European Centre for Medium-Range Weather Forecasts forecast $\mathbf{X}$. Auxiliary context variables $\mathbf{C}$ are mapped to a unified latent embedding via specialized encoders: periodic temporal coordinates are projected using learnable Fourier features (Li et al., 2021), while historical buoy measurements and tidal water levels are processed via 2D and 1D Convolutional Neural Networks, respectively. The encoder-decoder backbone, adapted from (Archambault et al., 2024) uses 3D-CNN along with Convolutional Block Attention Modules (Woo et al., 2018). The unified context embedding modulates the backbone via Feature-wise Linear Modulation (Perez et al., 2018). Finally, the decoder outputs the target wave buoy variance density spectrum.

$$\phi^* = \operatorname{argmin}_{\phi} \mathbb{E}_{(\mathbf{X}, \mathbf{C}, \mathbf{E}) \sim \mathcal{D}} [\mathcal{L}(g_\phi(\mathbf{X}, \mathbf{C}), \mathbf{E})] \quad (2)$$

where $\mathcal{L}$ is a loss function measuring the divergence between the predicted 1D spectra and the buoy-recorded ground truth $\mathbf{E}$. In practice, because the true underlying distribution is unknown, this expectation is approximated by the empirical risk calculated over the N available samples in a curated training set.

3.2. Deep Learning Architecture

3.2.1. Motivation

To model the complex mapping g_ϕ , we use a deep neural network, where ϕ represents the learnable weights of the architecture. The approach is motivated by the need to process high-dimensional, structured forecasts and capture the non-linear dependencies in wave energy distributions. Also, deep architectures naturally allow the fusion of multimodal context variables, for instance, using a homogeneous latent representation. Finally, it enables joint multidimensional estimation, in our case, energy across frequency and lead times, instead of requiring independent models. The proposed design for g_ϕ is described in the following section and schematized in Figure 5.

3.2.2. Backbone

The network backbone adapts the modular encoder-decoder architecture proposed by (Archambault et al., 2024), designed to efficiently process spatio-temporal data. The encoder takes $\mathbf{X}$ as input, and the decoder outputs $\hat{\mathbf{E}}$. Both use a custom gating module inspired by the Convolutional Block Attention Module (Woo et al., 2018). The spectral convolutions use circular padding along the directional dimension to account for the periodic geometry of the wave spectrum, and replicate (edge-value) padding along the lead-time and frequency dimensions. This avoids introducing artificial zero-energy values outside the retained frequency range.

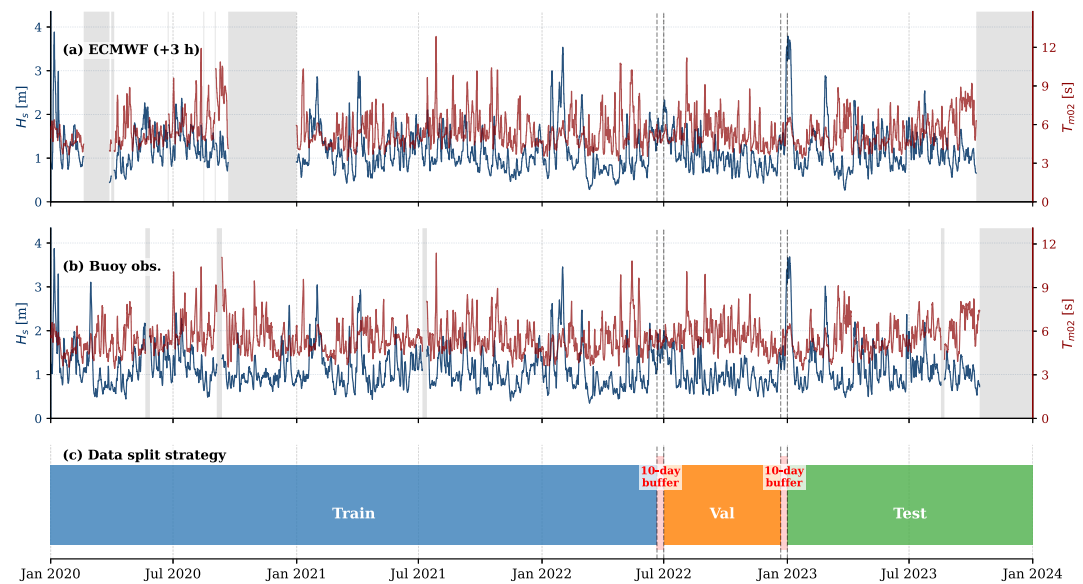


Figure 6. Temporal distribution and partitioning strategy. Evolution of significant wave height H_s and mean wave period T_{m02} for (a) European Centre for Medium-Range Weather Forecasts (ECMWF) forecasts and (b) buoy observations, with missing data shaded grey. (c) The chronological data split, with the holdout test set spanning January–October 2023 and strict 10-day buffers (red hatching) between training, validation, and test sets to prevent temporal data leakage. The gaps in the ECMWF record during 2020 result from incomplete local Meteorological Archival and Retrieval System retrievals and do not indicate interruptions of the operational forecasting system.

3.2.3. Context Modulation

The integration of context **C** is performed through a dedicated encoding and fusion framework. First, each variable is processed by a specialized encoder designed to map the raw data into a latent space of consistent dimensionality. Temporal coordinates are projected using learnable Fourier features (Li et al., 2021) to capture non-linear seasonal and diurnal periodicities (i.e., as seen in the buoy observations). Tidal water levels and the five most recent days of 1D spectral buoy measurements are encoded using 1D and 2D Convolutional Neural Networks, respectively. These individual representations are concatenated into a unified vector, providing a comprehensive descriptor of the environmental context. This fused context vector modulates the backbone in two possible ways. First, a cross-attention block enables the latent spectral representation to dynamically attend to the context vector. Second, Feature-wise Linear Modulation (FiLM) (Perez et al., 2018) performs context-dependent affine transformations on the activations of the encoder-decoder backbone.

3.2.4. Ensembling

To improve robustness and mitigate the variance inherent to random initialization, the final pipeline employs ensembling. The model is trained multiple times using different weight initializations. During inference, the final prediction is simply the average of all predictions. The spread across ensemble members can also be used to quantify the network's epistemic uncertainty.

3.3. Experimental Setup

3.3.1. Data Splitting

To ensure strict generalization, we partition the 3.5-year data set chronologically (Figure 6). The architecture is optimized over a 2.5-year window spanning from January 2020 to June 2022 (1,520 samples), whereas hyperparameters and early stopping criteria are tuned on a validation set ranging from July 2022 to December 2022 (368 samples). The final evaluation is performed on a holdout test set spanning January through October 2023. Furthermore, to mitigate temporal leakage between the 5-day historical context and the 5-day predictive horizon, explicit 10-day buffer zones are enforced at each partition boundary to prevent any overlap.

Table 1
Hyperparameter Search Space Explored During the Bayesian Optimization Process

Category	Hyperparameter	Search space
Architecture	Model capacity	{small, base , large}
	Cross-attention	{True, False }
Input	Modality	{ 2D spectra , 1D spectra}
	Scaling	{ True , False}
	Context: Buoy history	{True, False }
	Context: Periodic dates	{ True , False}
	Context: Local tides	{ True , False}
Output	Target formulation	{ direct , residual}
	Scaling	{ True , False}
Training	Loss formulation	{MSE, MAE , bin-weighted MSE, bin-weighted MAE}
	Learning rate	{ 10^{-3} , 5×10^{-4} , 10^{-4} }
	Batch size	{8, 16 , 32, 64}

Note. The optimal configuration, which minimized the validation H_5 RMSE, is indicated in bold. Note that while the final proposed architecture relies on 2D spectral inputs, 1D spectra were evaluated in the search space to quantify the predictive value of directional information.

3.3.2. Joint Optimization and Model Selection

To select a model and rigorously explore the design space, we deploy Bayesian optimization via the Tree-structured Parzen Estimator (Bergstra et al., 2013). The automated search evaluated 500 model configurations across 12 hyperparameters defining the optimization process, data processing, architectural design and context selection (see Table 1). Ultimately, the selected architecture is the one minimizing the validation Root Mean Square Error (RMSE) H_s , averaged uniformly across all forecast lead times.

3.3.2.1. Target and Loss Function

The optimization evaluates two fundamental target formulations: a direct prediction of the full wave spectrum, and a residual prediction designed to explicitly target the forecast errors. To quantify these errors, the optimization evaluates four distinct loss functions, generalized in Equation 3, where $p = 2$ yields the Mean Squared Error (MSE), $p = 1$ yields the Mean Absolute Error (MAE), and $w_f \in \{1, \Delta f\}$ toggles between the unweighted and frequency-bin-width-weighted counterparts. The bin-weighted variants integrate the error via a Riemann sum parameterized by the discrete frequency bin widths Δf , explicitly accounting for the non-uniform spacing of the wave spectrum. The validity mask is applied only to unavailable buoy observations. Valid spectral values are retained even when their variance density is close to zero. Therefore, physically low-variance-density boundary bins are not treated as missing data and continue to contribute to the loss.

$$\mathcal{L}_{p,w}(\phi) = \frac{1}{N} \sum_{i=1}^N \sum_{f=1}^{N_f} w_f |g_\phi(\mathbf{X}_i, \mathbf{C}_i)(f) - \mathbf{E}_i(f)|^p \quad (3)$$

Crucially, no explicit directional integration is performed within the loss function itself. The dimensional reduction from the 2D input space to the 1D frequency space is handled entirely by the internal architecture of the neural network g_ϕ . Consequently, the prediction $g_\phi(\mathbf{X}_i, \mathbf{C}_i)(f)$ evaluates directly to a 1D array that aligns dimensionally with the 1D buoy observation target tensor $\mathbf{E}(f)$.

3.3.2.2. Input Modality and Variable Selection

The search evaluates the importance of directional wave information by comparing models trained on the full two-dimensional directional forecast spectra as input with those trained on one-dimensional forecast spectra.

Additionally, we perform variable selection on auxiliary physical contexts to determine whether including tidal water levels, time/date, or historical buoy measurements improves the post-processing.

3.3.2.3. Architectural Capacity and Feature Fusion

The search explores the balance between representational power and the risk of overfitting by allowing varying backbone model sizes, multiplying or dividing the number of parameters by a factor of two. We also evaluate the usefulness of the cross-attention mechanism for context fusion, determining if simple FiLM is sufficient.

3.3.2.4. Optimization Dynamics and Standardization

The input and target spectra are optionally transformed element-wise according to

$$z = \frac{\log_{10}(E + \epsilon) - \mu}{\sigma}, \quad \epsilon = 10^{-8}, \quad (4)$$

where μ and σ are the corresponding statistics estimated from the training set. The loss (see Equation 3) is evaluated between the transformed prediction and target. Predictions are returned to physical space using

$$\hat{E} = \max(0, 10^{\mu + \sigma z} - \epsilon). \quad (5)$$

All bulk parameters and evaluation metrics are calculated from $\hat{E}$ in physical space. Model checkpoints are selected using validation H_s RMSE, also calculated in physical space. Network weights are updated using the AdamW optimizer (Kingma & Ba, 2014) combined with a linear warmup and cosine annealing learning rate scheduler. The learning rate of the optimizer and the batch size for the stochastic optimization are jointly searched with other hyperparameters.

4. Results

4.1. Selected Model

The model selection favored a simple configuration, excluding cross-attention and using MAE loss (Table 1). Regarding the input modality, the 2D spectra systematically outperform the 1D spectra, suggesting that the directional distribution constitutes relevant predictive information. As hypothesized in Section 2.3, including date encoding and tidal water levels as contextual information improves performance. Conversely, including the 5-day historical buoy sequence surprisingly degrades performance. Since these measurements are not assimilated into the operational ECMWF forecast, we expected them to be impactful. It is possible that the current architectural design limits the meaningful extraction of information from these observations. The selected model configuration and the optimization can be found in Table 1 and Figure 7.

4.2. Forecast Skill

Model performance is benchmarked against the ECMWF operational forecast using the holdout test set (January–October 2023) by evaluating integrated wave parameters derived from the 2D (ECMWF forecasted) and 1D (predicted) spectral variance density $E(f)$. These so-called bulk parameters are defined using the n^{th} -order spectral moment $m_n = \int f^n E(f) df$, yielding significant wave height ($H_s = 4\sqrt{m_0}$ [m]), mean period ($T_{m02} = \sqrt{m_0/m_2}$ [s]), and peak period ($T_p = 1/f_p$ [s]), where f_p is the frequency of maximum variance density. To diagnose the model's spectral correction capabilities across different wave regimes, the spectral integration is explicitly partitioned using a hard cut-off to distinguish between swell ($f < 0.1$ Hz) and wind sea ($f > 0.1$ Hz), with the threshold informed by the climatology shown in Figure 2.

4.2.1. Bulk Parameter Statistics

We observe a significant and persistent reduction in error within the swell partition. Table 2 details the RMSE for the extracted bulk parameters across the ECMWF baseline, the deterministic ML model, and the ML-Ensemble

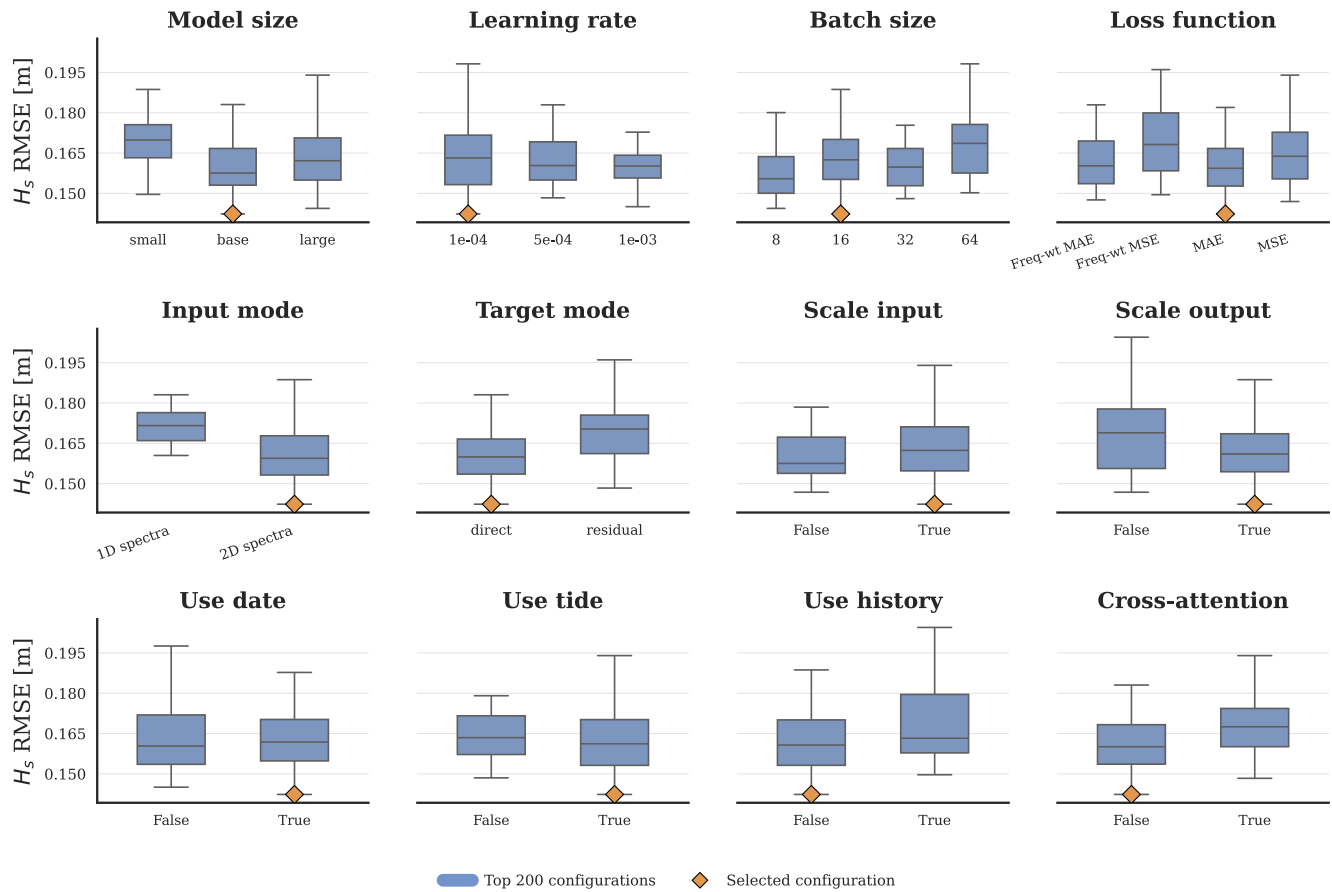


Figure 7. Hyper-parameters sensitivity: distribution of H_s Root Mean Square Error (RMSE), computed across the top 200 configurations evaluated during Bayesian optimization. Boxplots illustrate the distribution and interquartile range of the validation criteria (H_s RMSE averaged uniformly across all forecast lead times) for each parameter choice. The selected model is indicated with an orange marker.

across four forecast lead times (3, 24, 60 and 120 hr), highlighting these improvements. The ML-Ensemble systematically outperforms the baseline across all lead times for all swell derived bulk parameters. Most notably, the RMSE of the swell peak period (T_p) is reduced by approximately 37% at $t + 120$ hours and by up to 39% across the evaluated forecast lead times (Table 2). The proposed architecture also successfully corrects swell significant wave height (H_s), reducing RMSE by approximately 20%–27% across the evaluated forecast horizons. Conversely, performance in the wind sea partition is mixed. While the ML-Ensemble provides moderate improvements in wind sea periods, it fails to reliably track the energy estimation. Wind sea H_s RMSE is consistently degraded, which subsequently limits or degrades the total H_s performance (see Table 2). Finally, averaging predictions from different initializations reduces errors across most metrics. However, this ensembling may also smooth narrow or highly variable wind-sea peaks. Combined with the relative weighting induced by the logarithmic MAE, this may contribute to the conservative wind-sea estimates, although the contribution of each mechanism is not isolated here.

4.2.2. Bias and Variance Error Decomposition

Forecast errors can be decomposed into systematic (squared bias) and stochastic (variance) components. When examining the spectral density, ECMWF forecast errors are predominantly concentrated in the low-frequency swell and wind sea bands. While these errors mostly originate from the model incorrectly predicting the variance in wave energy (see Figure 8), a distinct systematic bias is also present. Similar swell biases have been reported for ERA5 by (Zieger, 2025), whose wave-model configuration did not include the explicit swell-attenuation term used in the operational forecasts considered here. In contrast, the ecWAM configuration used in this study includes the formulation of (Ardhuin et al., 2010). The remaining bias may therefore reflect its

Table 2

Root Mean Square Error (RMSE) Comparison of Wave Forecasting Models Across Total, Swell, and Wind Sea Components for Varying Forecast Lead Times ($t + 3$ hr to $t + 120$ hr)

Model	Total			Swell[0.04, 0.1]Hz			Wind Sea[0.1, 0.5]Hz		
	Hs[m]	Tp[s]	Tm02[s]	Hs	Tp	Tm02	Hs	Tp	Tm02
lead: $t + 3$ hr									
ECMWF	0.16	3.38	0.66	0.15	1.89	0.74	0.14	1.41	0.45
ML	0.21	2.93	0.62	0.12	1.48	0.56	0.19	1.31	0.43
ML-Ensemble	0.17	2.71	0.60	0.12	1.37	0.54	0.15	1.20	0.40
lead: $t + 24$ hr									
ECMWF	0.16	3.53	0.70	0.14	2.05	0.75	0.14	1.40	0.47
ML	0.19	3.24	0.63	0.11	1.44	0.53	0.18	1.39	0.43
ML-Ensemble	0.17	2.95	0.64	0.11	1.32	0.52	0.16	1.27	0.42
lead: $t + 60$ hr									
ECMWF	0.23	3.71	0.78	0.15	2.26	0.79	0.21	1.46	0.50
ML	0.23	3.13	0.69	0.11	1.45	0.55	0.22	1.35	0.46
ML-Ensemble	0.23	2.93	0.70	0.11	1.38	0.54	0.22	1.28	0.45
lead: $t + 120$ hr									
ECMWF	0.25	3.79	0.92	0.16	2.22	0.81	0.24	1.47	0.59
ML	0.25	3.33	0.87	0.13	1.55	0.58	0.25	1.41	0.55
ML-Ensemble	0.25	2.95	0.88	0.12	1.39	0.57	0.25	1.33	0.55

Note. Evaluated metrics include significant wave height (H_s), peak wave period (T_p), and mean period (T_{m02}). All reported RMSE values are expressed in their respective physical units. The ECMWF model serves as the performance baseline. Background shading for the ML and ML-Ensemble models denotes performance relative to the ECMWF baseline: green indicates a lower RMSE (improved performance), red indicates a higher RMSE (degraded performance), and grey denotes identical performance. For each metric and lead time, the lowest absolute error across all three models is highlighted in bold.

tuning, or issues in swell generation or propagation. Our model successfully removes the systematic swell bias and significantly attenuates error standard deviation across all frequencies and lead times. For higher frequencies associated with active wind seas ($f > 0.1$ Hz), the network introduces an additional negative bias, effectively under-predicting the amount of wave energy in this regime.

Integrating these spectral properties, we decompose the MSE of the significant wave height H_s to confirm that the baseline's total error is indeed dominated by stochastic variance (see Figure 9). The performance gains achieved by our model come almost exclusively from a significant reduction in forecast error variance in the swell frequency band. The baseline wind sea error remains largely unmitigated; the model struggles to resolve this stochasticity and instead augments the systematic bias.

4.2.3. Error Distribution in Frequency Space

Illustrated in Figure 10, we evaluate the signed error distributions for each frequency, $(\hat{E} - E)(f)$, aggregated across the test set at a forecast horizon of $t + 120$ hours. The ECMWF baseline systematically over-predicts spectral density in the frequency band ($0.08 \text{ Hz} < f < 0.1 \text{ Hz}$). Our correction largely neutralizes this effect; the median signed error is driven toward zero, and the error variance is substantially compressed. However, while correcting these deviations, our model ends up slightly under-predicting energy in the swell band.

This analysis is corroborated by the empirical Quantile-Quantile (Q-Q) distributions (Figure 11). The estimated swell significant wave height is much closer to the ground truth, yet slightly underestimated, while the ECMWF over-estimates swell H_s . Interestingly, the behavior of our model remains consistent across all forecast lead times, even for rarer events with elevated recorded H_s . However, the energy-reduction mechanism allowing swell correction propagates to higher frequency bands when the nature of ECMWF errors changes, resulting in systematic underestimation.

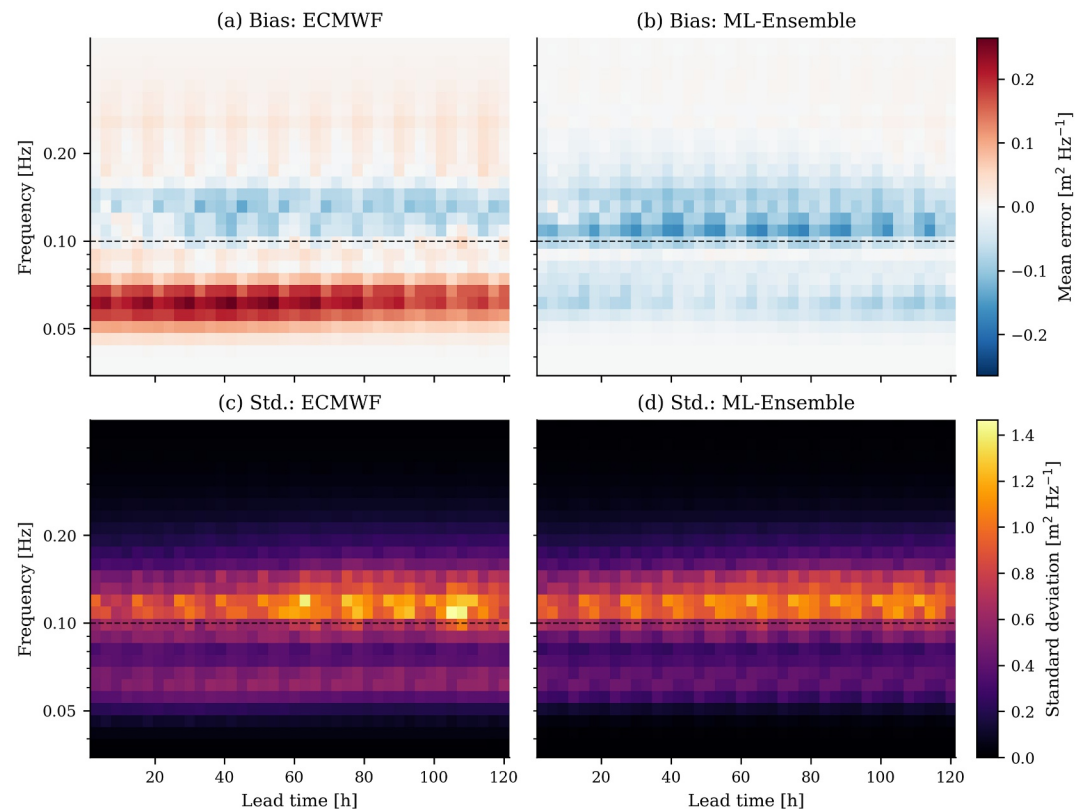


Figure 8. Heatmaps of frequency-lead time errors comparing the spectral energy prediction errors of the European Centre for Medium-Range Weather Forecasts baseline and the Machine Learning model. Bias and standard deviation are evaluated as a function of forecast lead time (x-axis) and wave frequency (y-axis). Globally bounded color scales are applied to ensure comparability of systematic and random error magnitudes across both models.

4.2.4. Frequency-Time Error Structure

4.2.4.1. Principal Component Analysis

To isolate the dominant spatiotemporal structures of the unadjusted ECMWF forecast errors and evaluate the efficacy of our post-processing methodology, we apply Principal Component Analysis to the joint time-frequency error matrix derived from the test set (Figure 12a). Out of 1,160 possible dimensions (40 lead times $\times$ 29 frequency bins), the first 10 principal components (PC or modes) account for 50.3% of the total baseline variance. While 10 modes are required to account for just over half the variance, out of the total number of orthogonal modes (1,160) this does provide evidence that the ECMWF errors are likely driven by underlying spatiotemporal structures, that can be learned and corrected, rather than random noise. However, the energy distribution across these modes indicates that there is no single time-frequency pattern resulting in the variance of ECMWF errors. While the overall error is heavily influenced by variance in the wind sea spectrum (consistent with Figure 8), the leading modes (PCs 1–7; Figure 12b) predominantly capture stationary variance within the swell band. Projecting the ML estimation onto these reference eigenvectors demonstrates significant reductions in error variance, particularly for modes with non-stationary swell components (PCs 8–10; Figure 12a), whose corresponding spatiotemporal structures are shown in Figure 12b. Specifically, the spatiotemporal structures of PCs 8 and 9 act together as a phase-shifted pair (see Figure 13a), distinguishing their behavior from the surrounding modes, such as PC 10 (Figure 12b), and highlighting arrival-time errors in the ECMWF forecasts of low-frequency swell.

4.2.4.2. Swell Arrival Time

Accurate prediction of swell arrival time remains a well-documented deficiency in numerical wave modeling (Jiang et al., 2016). Global forecast systems systematically exhibit a bias toward premature swell arrivals, and a

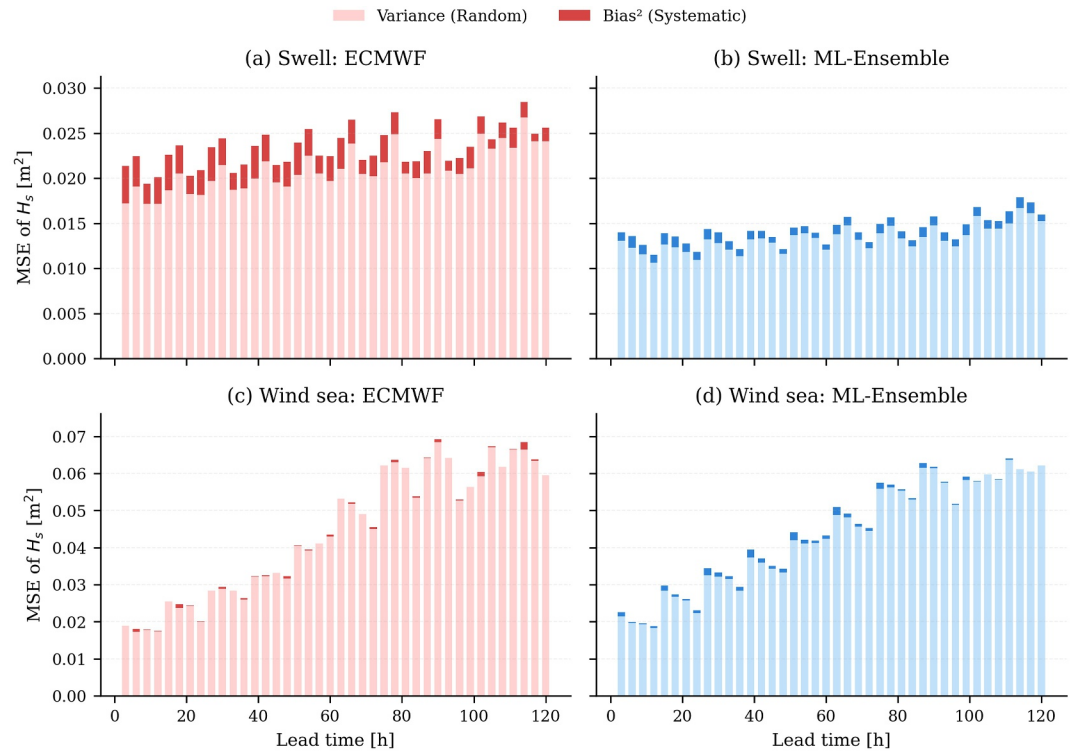


Figure 9. Mean Squared Error (MSE) decomposition for significant wave height (H_s), segmented by physical sea state components using a frequency cut-off of 0.1 Hz to distinguish between Swell ($f < 0.1$ Hz) and Wind Sea ($f > 0.1$ Hz). The bars represent total MSE, partitioned into systematic squared bias (darker shade) and random variance (lighter shade). This breakdown illustrates the Machine Learning model's efficacy in mitigating both the persistent bias and the stochastic error components inherent in the European Centre for Medium-Range Weather Forecasts baseline across distinct spectral regimes.

recent study (Sun et al., 2026) confirms an early-arrival bias in the Indian Ocean and along the NWS of Australia, relatively close to our location of interest. As noted, Modes 8 and 9 partially capture these arrival-time mismatches (Figure 12b). By computing the difference between PC_8 and PC_9 (Figure 13a), we compose an error type explicitly tracing a dispersive time shift. To quantify the mitigation of these propagation errors, we isolate test set samples where the principal coefficients PC_8 and PC_9 have opposite signs. As demonstrated in Figures 13b and

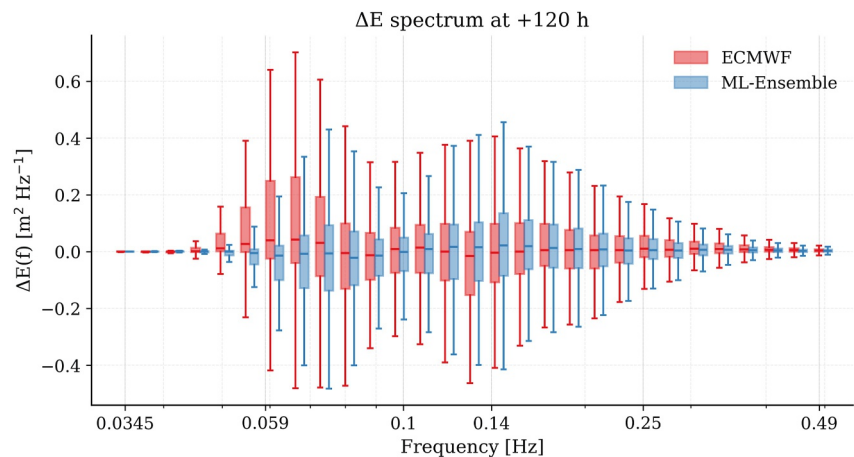


Figure 10. Boxplots of signed spectral errors ($\hat{E} - E$)(f) at a 120 h lead time, evaluated across discrete frequency bins. The ML-Ensemble systematically compresses the error variance and eliminates the pronounced positive bias present in the European Centre for Medium-Range Weather Forecasts baseline's primary swell band.

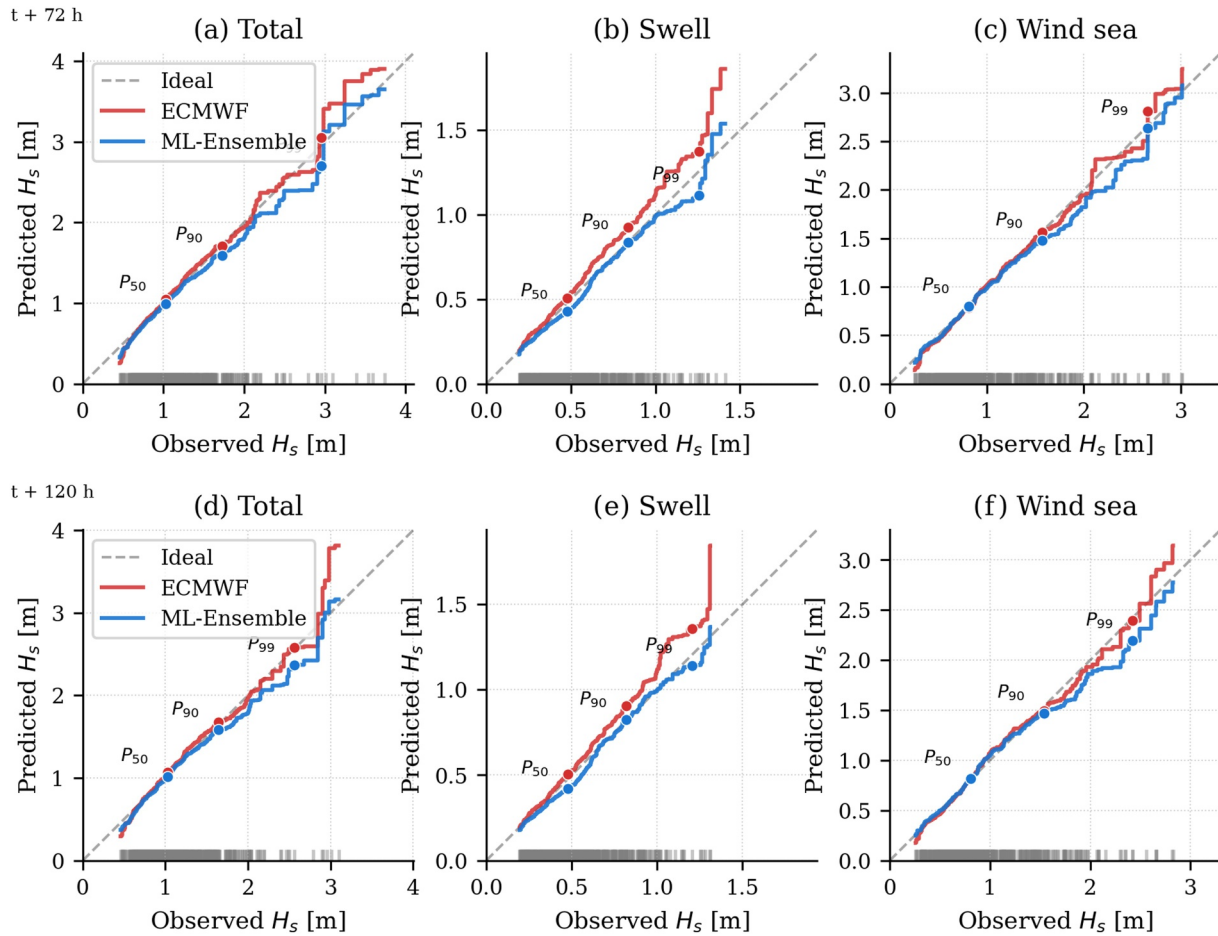


Figure 11. Empirical Quantile-Quantile (Q-Q) distributions for Significant Wave Height (H_s) at lead times $t + 72$ h (top) and $t + 120$ h (bottom), partitioned into swell and wind sea components.

13c, our model systematically reduces the magnitude of these dispersive errors ($PC_8^2 + PC_9^2$). We argue that this energy reduction is an informative proxy indicating the correction of swell arrival times. To illustrate our point, we select the ECMWF Forecast with the highest energy in errors ($PC_8^2 + PC_9^2$). In this specific event (Figure 14), the ECMWF exhibits the typical early-arrival issue discussed in (Sun et al., 2026). Our ML correction correctly estimates the swell arrival time, more than 60 hr after the ECMWF forecasted arrival time.

4.3. Ensemble Spread

A benefit of the ensemble approach we have employed is that it allows an estimate of the uncertainty in the post-processing, specifically the model's epistemic error. To evaluate the probabilistic predictive capacity of the ML-Ensemble, we compute stratified rank histograms (Talagrand, 1999). As shown in Figure 15, the resulting distributions exhibit a persistent U-shape across all evaluated parameters and lead times. In a perfectly calibrated ensemble, this histogram would be flat. The observed U-shape is consistent with substantial under-dispersion: the ensemble spread is too narrow relative to the verifying buoy estimates, which frequently fall either above the highest ensemble member or below the lowest. However, rank histograms are also sensitive to observation error (Saetra et al., 2004). Sampling uncertainty in the 30-min buoy records and representativeness differences between the point measurement and the ECMWF grid cell may therefore contribute to the departure from a uniform distribution. This lack of spread occurs because our ensemble members are optimized using deterministic loss functions and differ from one another solely through random weight initialization. Consequently, the resulting spread only captures a narrow band of the model's internal parametric doubt (epistemic uncertainty). It fails to represent the inherent physical randomness and chaotic variability of the wave forecast errors themselves

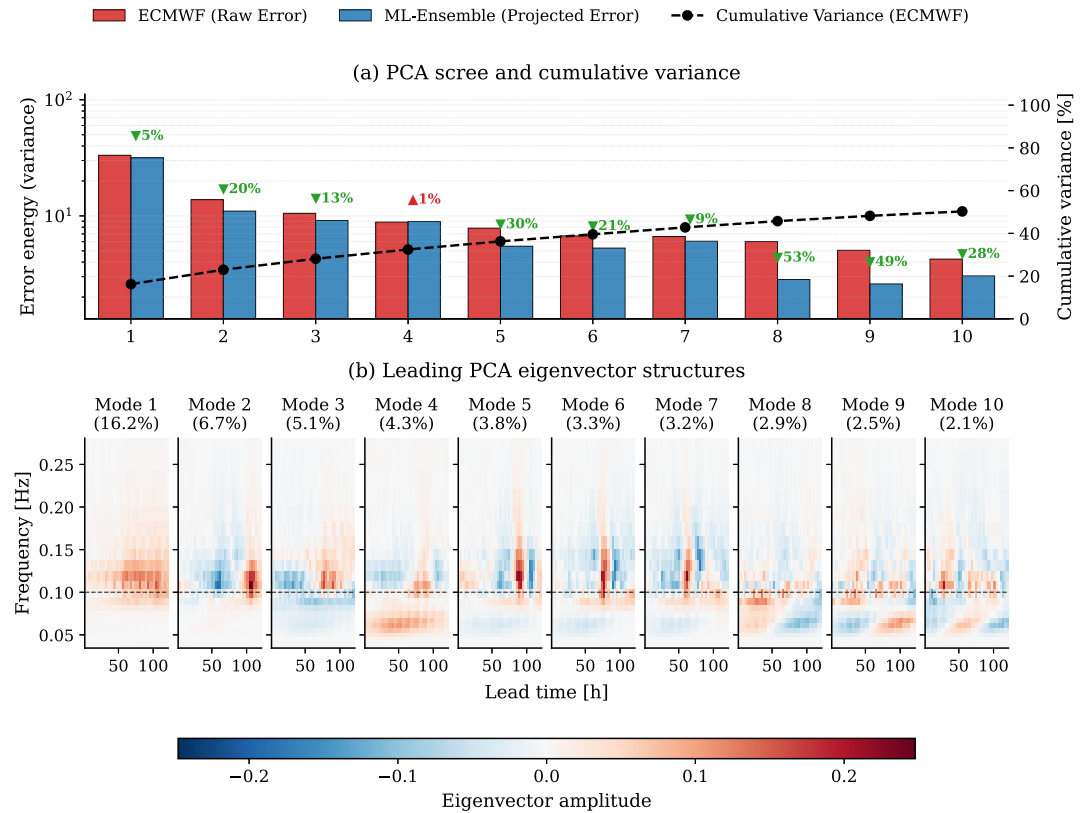


Figure 12. Patterns of frequency–lead-time error via Principal Component Analysis. (a) Error variance associated with the first ten ECMWF-derived principal components for the European Centre for Medium-Range Weather Forecasts (ECMWF) baseline and the ML-Ensemble errors projected onto the same eigenvectors; the dashed line shows the cumulative variance explained by the ECMWF modes. (b) Frequency–lead-time structures of the first ten ECMWF-derived eigenvectors, with the percentage of total ECMWF error variance explained by each mode shown above each panel.

(aleatoric uncertainty). While the ensemble effectively regularizes deterministic predictions by averaging, it does not presently provide a calibrated probabilistic forecast; the apparent under-dispersion reflects both the limited ensemble construction and, potentially, uncertainty in the verifying observations.

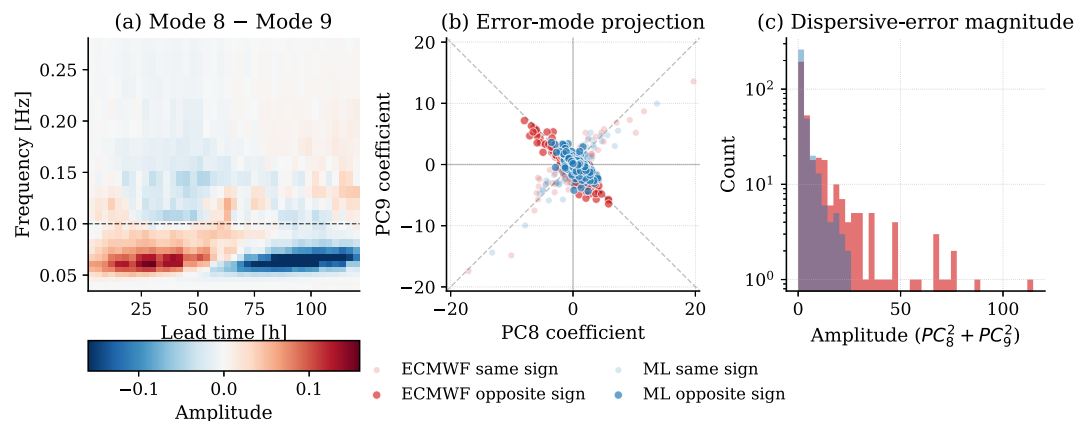


Figure 13. Swell arrival time error pattern. (a) Spectral structure of the combined principal error modes (Mode 8 minus Mode 9), characteristic of swell arrival time mismatches in the frequency–lead time domain. (b) Projection of forecast errors onto the Mode 8 and Mode 9 phase space. Opaque markers highlight extreme opposite-sign phase shifts. (c) Distribution of error magnitudes ($PC_8^2 + PC_9^2$) for the opposite-sign subset.

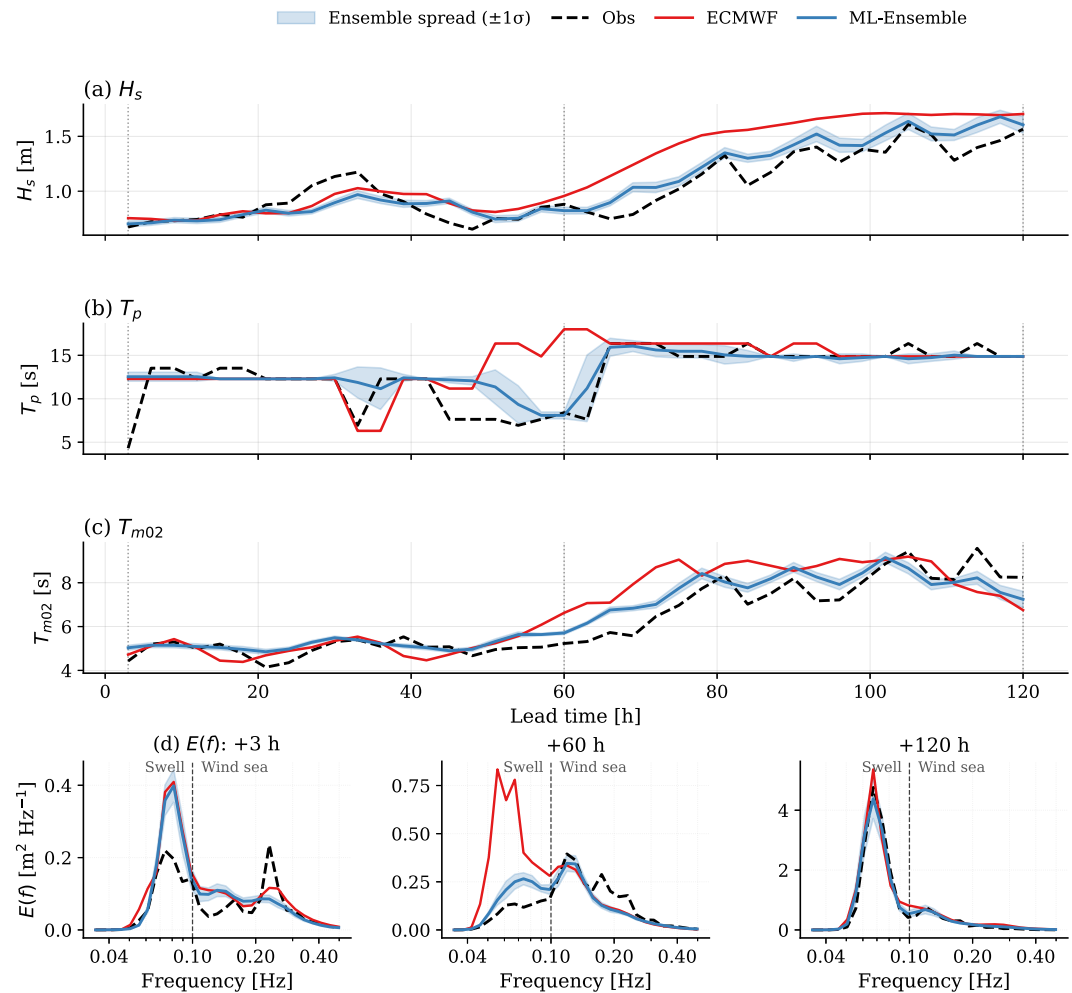


Figure 14. Case study of an early swell arrival forecast from the European Centre for Medium-Range Weather Forecasts (ECMWF) baseline (issued on 2023-09-11 at 12:00 UTC). The ML-Ensemble corrects the ECMWF premature energy peak, successfully realigning the predicted arrival time with empirical observations. Note the change in vertical scale between the 1D spectral panels. The displayed spectra remain smooth at the lowest and highest retained frequencies, with no artificial peaks introduced at the spectral cutoffs.

5. Conclusion

5.1. Summary

In this work, we formulate a site-specific post-processing of spectral numerical wave forecasts as a supervised DL regression task. We map the numerically forecasted 2D spectra time series, to the 1D spectra recorded by a wave buoy, including environmental context captured via tidal water levels and the time as a means of incorporating seasonality and other temporal variability. We show that the proposed model can meaningfully correct ECMWF baseline predictions. The improvements are mainly concentrated in the swell frequency band (frequencies < 0.1 Hz), reducing average errors in bulk parameters H_s , T_p and T_{m02} in the swell band. The model also demonstrates skill in adjusting the arrival time of swell events.

5.2. Limitations

5.2.1. Data Accessibility and Record Length

While our results demonstrate significant improvements in local forecast skill, the methodology was trained and evaluated on a relatively short temporal record (3.5 years). This limited duration restricts the model's ability to fully capture and learn from broader inter-annual climate variability, multi-year seasonality cycles (such as the El

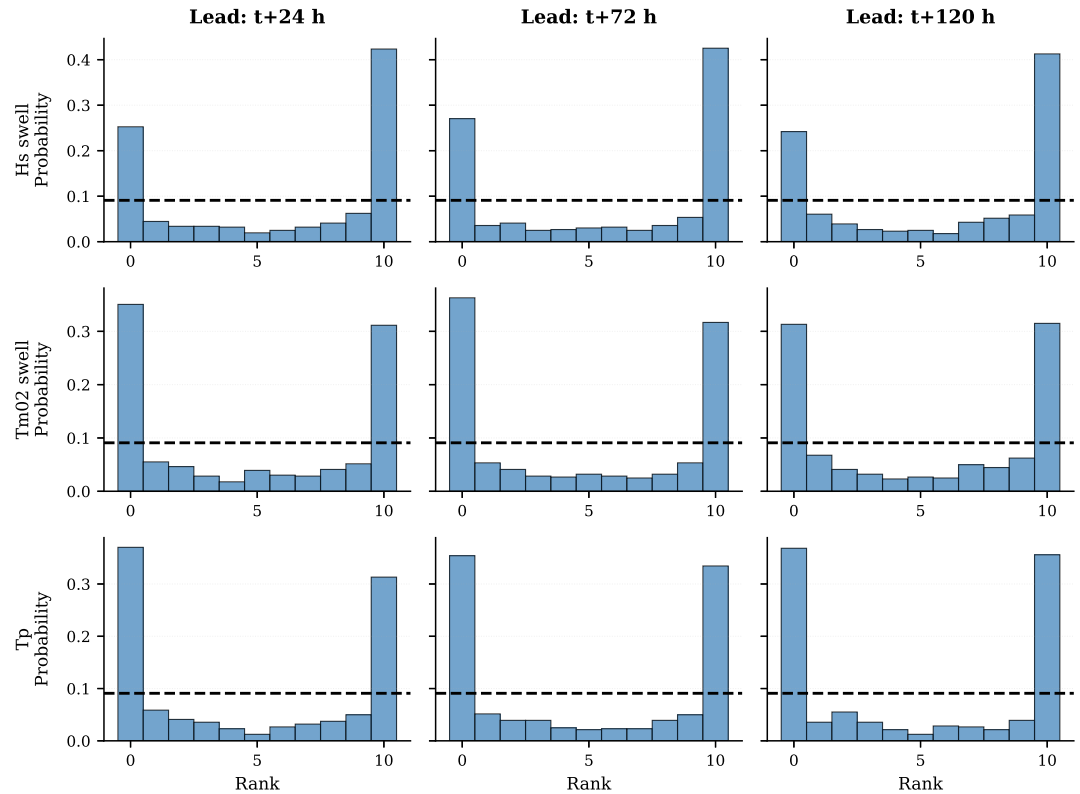


Figure 15. Stratified rank histograms (Talagrand diagrams) evaluating the probabilistic calibration of the ML-Ensemble for swell Significant Wave Height (H_s) and Mean Zero-Crossing Period (T_{m02}). The persistent U-shape across all forecast horizons indicates under-dispersion.

Niño-Southern Oscillation or the Indian Ocean Dipole), or rare extreme events. The primary bottleneck constraining the length of this training set is the time required to obtain archived operational forecasts. Compiling this continuous 3.5-year dataset required nearly 2 years of continuous query execution from the ECMWF MARS. Consequently, these practical limitations regarding data accessibility restricted our experiments.

5.2.2. Evolving Nature of the Data Set

The operational ECMWF forecasting system is periodically updated (ECMWF, 2023; ECMWF, 2026), introducing discrete shifts in the baseline forecast skill resulting from changes in the model settings, computational grid and resolution, and physical process representations within the ecWAM model. For instance, Cycle 47r1 of the forecasts, which became operational in 2020, effectively capped the surface drag coefficient under strong winds. This non-physical threshold was later replaced in 2024 in cycle 49r1. These structural updates, combined with possible buoy sensor replacements, introduce temporal inconsistencies into the training set that violate the assumption of stationarity and potentially bias the learned correction.

5.2.3. Unexploited Local Observations

The operational ECMWF wave forecast from 2020 onward (Cycle 47r1) assimilates H_s from satellite altimeter missions, but not local buoy observations. Our expectation was that including a 5-day historical buoy sequence would capture these unassimilated local persistence and improve short-term forecasting. However, in our study, incorporating a 5-day historical buoy sequence degraded forecast skill and was explicitly rejected during hyperparameter optimization. The current architecture failed to condition the correction on recent observations.

5.2.4. Omitted Atmospheric Wind Context

The absence of explicit wind forcing may partly explain the limited performance in the wind-sea regime. Although the ecWAM spectra already reflect the response of the wave model to the forecast winds, supplying

lead-time-matched wind speed and direction would allow the post-processing model to condition its correction directly on local atmospheric forcing. A separate multi-year atmospheric archive was not assembled because the retrieval and processing of the spectral dataset already represented a substantial data-acquisition effort within the available project timeframe. However, incorporating such context would be straightforward using the proposed framework.

5.2.5. Uncertainty Quantification

As demonstrated by the rank histograms, our ensemble is of no use when it comes to quantifying uncertainty. Because the architecture is optimized via deterministic regression, it captures a narrow band of epistemic uncertainty. Reliable uncertainty quantification will require transitioning to explicit generative modeling.

5.3. Improvement Perspectives

5.3.1. Exploitation

5.3.1.1. Directional Observation

In the current formulation, the target variable is restricted to the 1D frequency variance density $E(f)$ spectrum. However, most wave buoys inherently capture the directional spreading of wave energy, typically parameterized by the directional Fourier coefficients (a_1, b_1, a_2, b_2) . Future iterations of the model could expand the regression target space to predict these coefficients alongside the 1D spectrum. Previous approaches, for example (Hansen et al., 2022), have aimed to directly correct the estimated model 2D spectrum. However, single point directional wave buoys do not inherently measure the 2D spectrum with this rather being a statistically modeled quantity using the 1D and directional moments as inputs (e.g., via the Maximum Entropy Method (Lygre & Krogstad, 1986)). As this step adds additional uncertainty in the measurement space, our focus was on adjusting the 1D spectrum from which the most utilized wave bulk parameters are derived.

5.3.1.2. Super-Resolving in Frequency and Time

The pre-processing methodology interpolated the higher-resolution wave buoy 1D spectrum (100 frequency bins) to the coarser ECMWF frequency grid (29 bins). Because DL architectures naturally accommodate dimensionality mappings, the regressor could be designed to inherently super-resolve the 29-bin ECMWF input directly into the 100-bin target resolution. The same remark is valid for time granularity, since buoy measurements are available every 30 min.

5.3.2. Exploration

5.3.2.1. Generalizing the Correction in Space

Currently, the model is specialized for one specific site and would have to be retrained to be applied to another location. With a suitable multi-site dataset, spatial generalization should be assessed through cross-station validation and zero-shot testing at neighboring locations. Such experiments would help determine whether the learned correction captures a basin-wide systematic ECMWF error, particularly the premature swell-arrival bias, or predominantly site-specific effects associated with local bathymetry and measurement conditions. Assuming there is a spatial coherence and dependency in the ECMWF forecast error, and so the sought correction, it should be possible to learn over multiple sites in one training cycle, encoding the geo-location information. For instance, leveraging a model like DeepONet (Lu et al., 2021) would theoretically provide corrections at arbitrary spatial coordinates, even if they are not present in the dataset. Such an ambitious model would raise familiar concerns encountered in data assimilation, such as cross-calibration, since different directional wave measurements may vary widely with the buoy model (Ardhuin et al., 2019).

5.3.2.2. Regime-Dependent Modeling

A monolithic network may struggle to balance the correction dynamics across both calm and extreme, or wind sea and swell wave, conditions or both. A practical solution is regime-dependent modeling (McCandless et al., 2016). This approach first groups the input data into distinct wave states, and a separate network is then trained for each

specific group. However, it also introduces new challenges such as regime identification and data set fragmentation.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The ECMWF spectral wave forecast data were retrieved from the ECMWF Meteorological Archival and Retrieval System (MARS) (European Centre for Medium-Range Weather Forecasts, 2026), following the ECMWF data policy. It took us approximately 2 years of continuous downloading to gather 3.5 years of history. This long acquisition time reflects limitations in retrieval efficiency rather than unavailability of the archived data. The missing ECMWF segments visible in 2020 reflect incomplete local archive retrievals caused by interrupted MARS requests, rather than periods when the operational forecasting system was unavailable. Wave buoy measurements were provided by Shell Australia. Due to commercial data restrictions, this data set is not publicly available. Researchers interested in accessing the data for replication purposes may submit a formal request to Senior Metocean Engineer, Shell Australia. Prior to analysis, the raw binary files were retrieved from the buoys and processed using the Datawell `libDatawell` software library (Datawell BV, n.d.). The machine learning pipeline and supporting analysis code are archived on Zenodo (Filоче et al., 2026).

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